

IoT-Enabled Smart Door Lock Featuring RFID, OTP and Intrusion Detection

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Abstract — The integration of IoT into modern security systems has significantly improved the access control and monitoring capabilities. This paper presents the design of a smart door lock system that combines the radio frequency identification (RFID), one-time password (OTP), and intrusion detection capabilities using a number of sensors, actuators and cloud-based platform. The proposed system aims to improve residential and institutional security by enabling a multi-factor authentication and real-time breach alerts. The authorized user can unlock the door using RFID tag while an OTP is generated and verified via a cloud server in case of an invalid RFID scan. At the same time, intrusions are detected through motion sensors and alerts are sent immediately to the admin via IoT enabled notifications. The testing results include the authentication success rate of 97%, intrusion detection accuracy of 90%, cloud uptime of 98.5%, RFID latency of 0.8 seconds and OTP latency of 1.2 seconds. The proposed system can be deployed to implement a secure and user-friendly door access in the offices, residential buildings, hostels and academic institutions, etc.

Keywords— *Internet of Things (IoT), Smart Lock, OTP, RFID, Intrusion Detection.*

I. INTRODUCTION

With the rising need for smarter security solutions in both residential and commercial environments, conventional lock-and-key mechanisms are increasingly proving to be vulnerable, inconvenient, and outdated. Burglary cases, unauthorized entries, and key misplacements highlight the critical need for advanced access control systems. A smart door locking system offers a highly secure and automated alternative that not only enhances physical safety but also integrates seamlessly with digital technologies [1-4]. Among these, the combination of RFID, OTP-based verification, and real-time alert systems stands out as a robust multi-layered defence mechanism. The integration of radio frequency identification enables contactless identification through embedded microchips within cards or tags. These RFID tags are uniquely coded and act as the first level of authentication, replacing physical keys with programmable digital identifiers. This not only enhances user convenience but also reduces the possibility of unauthorized duplication, as only registered tags are recognized by the system [5-8]. The RFID module acts as a gatekeeper, initiating the second security phase only upon successful validation. To further strengthen access control, the system incorporates an OTP mechanism via GSM communication. After an RFID tag is verified, a unique 6-digit OTP is generated and sent to the user's registered mobile number using GSM module. This additional step ensures that

even if a tag is lost or cloned, entry cannot be gained without simultaneous mobile verification. OTPs expire after a short duration and cannot be reused, ensuring dynamic, time-bound security and eliminating the risk associated with static passwords or pins. In case of invalid RFID tags or incorrect OTP entries, the system activates a built-in alert mechanism that serves both as a deterrent and a notification system. A loud buzzer is triggered immediately to draw attention, while a warning message is simultaneously sent to the administrator's mobile number. This feature ensures that any unauthorized attempt is documented in real time, providing situational awareness even when the user is away from the physical premises [9-12]. This work aims to develop and experimentally validate a multi-factor smart locking system that can be applied in homes, office cabins, secure storage units, and other restricted zones. By combining RFID, GSM-based OTP, and alert notifications, the proposed system not only improves access control but also enhances reliability and trust in smart security systems. This layered security approach ensures that even if one layer is compromised, the others continue to protect the integrity of the access system, making it a powerful and scalable solution for modern-day security challenges [13-14]. Beyond home and office applications, the proposed system is particularly beneficial in environments that demand restricted access to sensitive areas such as laboratories, server rooms, and vaults. Traditional lock systems, even if reinforced, are often rendered ineffective without audit trails or real-time breach alerts. The integration of an IoT-based architecture allows the proposed smart door lock to be remotely monitored, controlled, and logged ensuring complete visibility into entry events. This means administrators can retrieve data on access history, failed attempts, and even activate lock or alert states remotely through cloud-connected modules if extended in future versions. Moreover, the modular nature of this system ensures scalability and upgradability. Additional layers such as biometric authentication or mobile app integration can be seamlessly incorporated into the existing setup. This adaptability future-proofs the design, aligning it with the ongoing advancements in smart home automation and cybersecurity [15]. The implementation also emphasizes affordability, using readily available components like Arduino Uno, RFID, GSM, and an LCD display making it feasible for widespread adoption without compromising on core security features. In addition to enhancing physical security, the proposed system also addresses one of the key challenges in digital access control and user authentication management. Unlike traditional keys or static passwords, RFID tags can be issued, revoked, or updated without physically altering the

lock mechanism. This flexibility makes the system especially useful for institutions or shared accommodations where user turnover is frequent. Moreover, OTP-based verification adds a second layer that is personalized and non-transferable, as the OTP is sent only to the registered phone number, thus ensuring that access is granted only to the intended individual. The proposed work serves as an excellent example of how embedded systems, wireless communication, and real-time alert mechanisms can be combined to create practical and impactful solutions. It also encourages learners to engage with interdisciplinary technologies such as microcontroller programming, GSM networking, RFID data handling, and user interface design. By implementing real-world challenges in the field of smart security, this work not only contributes a viable solution but also opens the door to further research and innovation in the field of IoT and smart automation sector.

II. ARCHITECTURE OF THE SMART DOOR LOCK SYSTEM

The architecture of the proposed smart door lock system is structured to deliver a secure, multi-factor authentication and real-time intrusion monitoring using low-cost hardware and cloud-based IoT integration. The design incorporates hardware modules simulated on the Wokwi platform and cloud communication using the Blynk IoT service. This section outlines the core architectural components, data flow, hardware-software interaction, and the logical integration of OTP verification, RFID authentication, and intrusion detection mechanism.

A. Overview

The architecture of the proposed system has been presented in Fig.1. It can be logically divided into three primary modules: authentication unit, intrusion detection unit, and cloud communication and notification unit. These modules are controlled centrally via an ESP32 microcontroller, which is programmed using the Arduino IDE and simulated on Wokwi platform for functional validation. Each unit is integrated with corresponding hardware and software components, ensuring seamless operation across multiple layers of the security system. The overall workflow begins with a user attempting to access the door. The system first checks for a valid RFID card. If unavailable or denied, the system generates a one-time password sent through Blynk cloud platform. Concurrently, the system constantly monitors the premises for unauthorized movement using an ultrasonic sensor and triggers a real-time alert in case of intrusion. All access attempts, authentication statuses, and alerts are logged and communicated through the Blynk Cloud-platform.

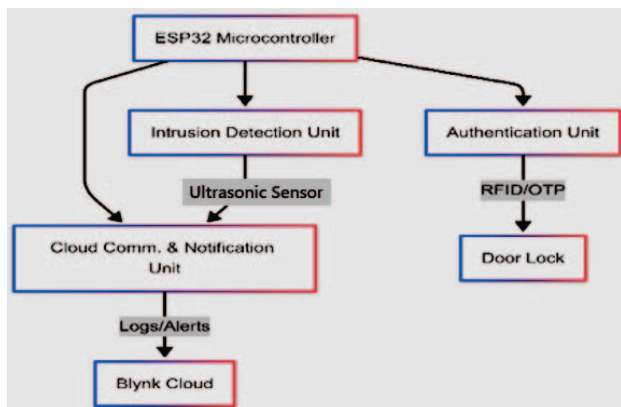


Fig.1: Architecture of the proposed IoT-based door lock system.

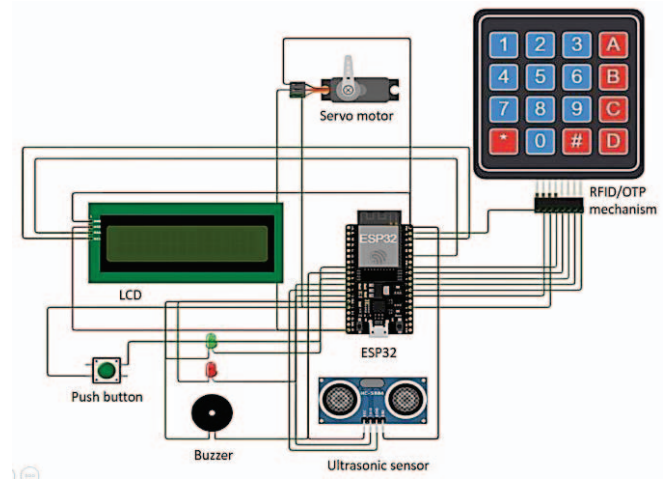


Fig.2: Hardware circuit diagram of the proposed system.

B. Hardware Components and Configuration

The hardware circuit diagram of the proposed system has been depicted in Fig.2. It consists of the following components.

ESP32 microcontroller: The ESP32 serves as the brain of the system. It is a Wi-Fi-enabled microcontroller that handles sensor input, actuator control, and cloud communication. Its dual-core performance allows for parallel handling of authentication and intrusion monitoring without significant delays.

RFID module (RC522): The RC522 RFID reader is used to scan authorized RFID tags. When a tag is presented near the reader, the ESP32 reads its UID (unique identifier) and compares it to a predefined list of authorized IDs stored in the firmware.

Ultrasonic sensor: The ultrasonic sensor detects movement within a specified range. When motion is detected while the system is in a secured state, it is treated as a potential intrusion. The sensor's output is monitored continuously by the ESP32, and intrusion events are logged.

Servo motor: A micro servo motor is connected to the door lock mechanism. Upon successful authentication via RFID or OTP, the ESP32 sends a signal to the servo to rotate to the unlocked position and allowing the physical access. After a timeout of 5 seconds, the servo returns to the locked state automatically.

Buzzer: A buzzer is triggered to provide audible feedback in cases of failed authentication or detected intrusions, alerting nearby occupants in real-time.

LCD: After the RFID or OTP verification process, the status of the access has been displayed on an LCD.

Wokwi Simulation Environment: All components are simulated in the Wokwi simulator, allowing for complete virtual prototyping. The simulation replicates real-world timing and logic, making it ideal for verifying functionality without deploying physical hardware.

C. Cloud Communication and Blynk Integration

The system uses the Blynk IoT platform to enable cloud-based communication and user interaction. The Blynk platform provides a customizable interface connected to virtual pins on the ESP32. This integration supports OTP generation and display via Blynk text widget, user input for OTP verification via terminal and real-time notification alerts

for intrusion detection using Blynk's notification API. When a user requests access via OTP, a six-digit random code is generated on the ESP32 and sent to the Blynk app. Upon entering the correct code, the ESP32 verifies the input and unlocks the door. If an incorrect code is entered, the access is denied and the user is prompted again. In case of intrusion, the ESP32 pushes a notification to the Blynk app, alerting the user regardless of their physical proximity to the premises. This ensures remote monitoring and control capability.

D. Authentication Flow Logic

The system prioritizes RFID authentication as the primary access method for registered users. The logic flow is as follows:

1. Detection of RFID Scan:
 - ESP32 reads RFID tag's UID.
 - If UID matches authorized list, then unlock the door via servo.
 - If UID is unrecognized, then prompt user to request OTP.
2. OTP Request:
 - A virtual button in the Blynk App triggers the OTP generation.
 - ESP32 generates a time-sensitive OTP and pushes it to Blynk.
 - User enters OTP via the app.
 - If OTP is valid, unlock door; else deny access and log the event.

This two-layered mechanism provides both convenience and security by supporting fallback authentication and tracking the unauthorized attempts.

E. Intrusion Detection and Alert Flow

The intrusion detection unit remains always active. It operates independently from the authentication unit to monitor for motion during idle or locked states. The logic flow is as follows:

- When the ultrasonic sensor detects motion, ESP32 controller confirms the movement and checks if system is in "locked state."
- If found active, the buzzer is triggered and Blynk platform based alert is sent.
- Motion logs are stored for the review.

This real-time monitoring capability significantly improves the security quotient by alerting users about potential break-ins or unauthorized presence even without attempted access.

F. Functions of Wokwi and Blynk platforms

The choice of Wokwi for simulation and Blynk for IoT communication provides several advantages. Wokwi allows for rapid prototyping and debugging of complex microcontroller logic without hardware costs. It offers real-time serial monitoring, virtual devices (RFID, ultrasonic sensor, servo), and accurate timing simulations. Blynk simplifies cloud communication through a no-code mobile interface and seamless ESP32 integration. Its notification, widget, and virtual pin systems allow flexible and scalable interaction without the need to build a custom mobile app. Together, these platforms enable complete validation of the smart door lock architecture without deploying physical infrastructure, making the system highly adaptable and cost-effective during the development phase.

III. IMPLEMENTATION METHODOLOGY

The functional overview of the proposed system is shown in Fig.3 where the user starts the process with an RFID card scan. If the card is valid, the door gets unlocked, otherwise the user is prompted for an OTP input. Again, if the OTP is valid, the doors is unlocked, else the access is denied. An ultrasonic sensor has been used to trigger an alert of the coming user as well. Upon detection of a user, the buzzer sounds up with alert messages on the LCD.

The Fig.4 depicts the RFID-based access control process for the proposed system. The reader module RC522 reads the user's RFID card and send data to ESP32 which further validates it based on the matching with stored database. If a match has been detected, the servo motor mechanism opens the door, else the access is denied.

The OTP verification functionality has been presented in Fig.5. If the RFID card is not recognized, the registered member will be prompted to input an OTP. The grant will be accessed to open the door upon verification of the OTP. If wrong OTP is inserted, the access will be denied.

The intrusion detection mechanism has been presented in Fig.6. Once the ultrasonic sensor detects a person, the programmed logic will check for any recent RFID scan or an OTP prompt. For an invalid RFID or OTP, the buzzer and LCD have been activated to indicate the presence of intruder.

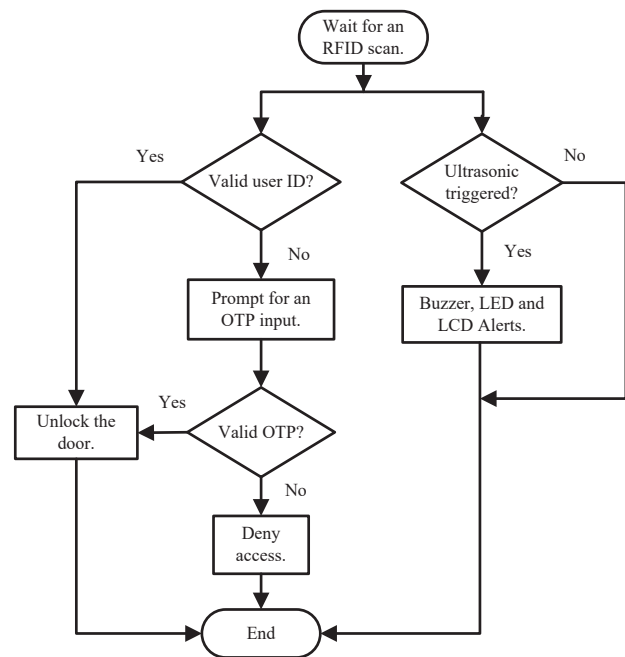


Fig.3: Functional overview of proposed system.

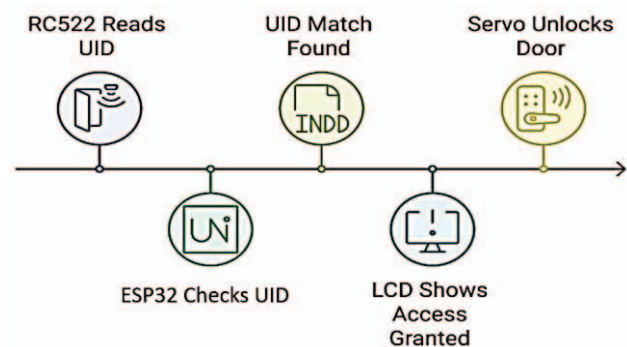


Fig.4: RFID-based access control.

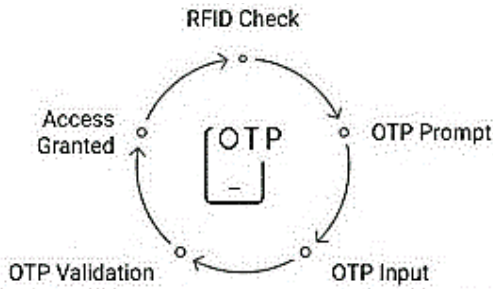


Fig.5: OTP-based access control.

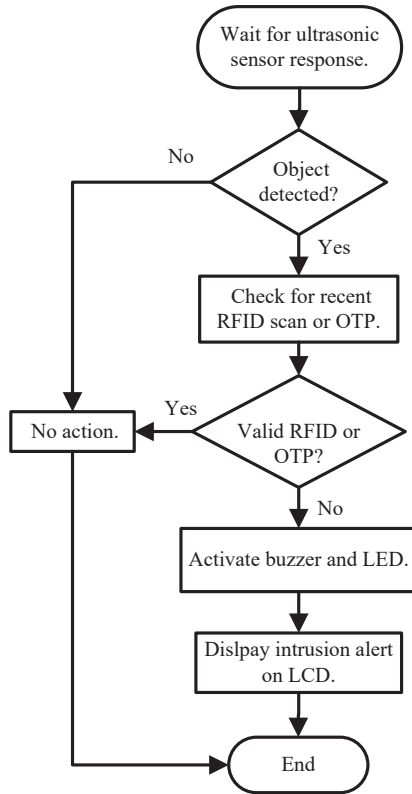


Fig.6: Intrusion detection mechanism.

IV. RESULTS AND DISCUSSION

The performance of the proposed smart door lock system has been evaluated using a simulation-based testing environment through the Wokwi platform, with real-time cloud communication facilitated via Blynk IoT application. This section presents the evaluation criteria, methodology, and results across various system metrics including authentication reliability, intrusion detection accuracy, system latency and the OTP responsiveness.

A. Authentication Success Rate

Fig.7 presents the authentication success rates of the proposed system under RFID and OTP testing scenarios. To evaluate the reliability of the dual authentication system, 100 access attempts have been simulated. Fifty attempts used valid RFID tags, and the remaining used OTP-based access. The system correctly granted access in 49 of the 50 RFID scenarios and 48 of the 50 OTP cases. The combined authentication success rate has been recorded as 97%, demonstrating a high level of reliability in verifying authorized users.

B. Intrusion Detection Accuracy

Intrusion detection capability has been evaluated by using the ultrasonic sensor integrated into the system. A number of controlled tests have been performed by introducing both genuine intrusion scenarios and normal motion cases. The ultrasonic sensor has been tested 40 times out of which 20 involved unauthorized presences, and the rest involved normal conditions such as background motion or environmental changes. The system correctly detected 19 intrusions (true positives) and misclassified 2 normal events as the false positives. From this data, the precision and recall performance has been evaluated as:

- **Precision** = $19 / (19 + 2) = 90.47\%$
- **Recall** = $19 / (19 + 1) = 95\%$

Fig.8 depicts the precision of recall performance of the proposed system for intrusion detections. These findings suggest that the system is highly sensitive to unauthorized activity while maintaining reasonable resistance to the false alarms.

C. OTP Delivery and Verification Time

The OTP feature has been tested for responsiveness under different network conditions. On an average, OTPs have been received within 2.1 seconds via the Blynk app, and users have taken approximately 5–7 seconds to enter and verify the code. The total time taken from OTP generation to successful door

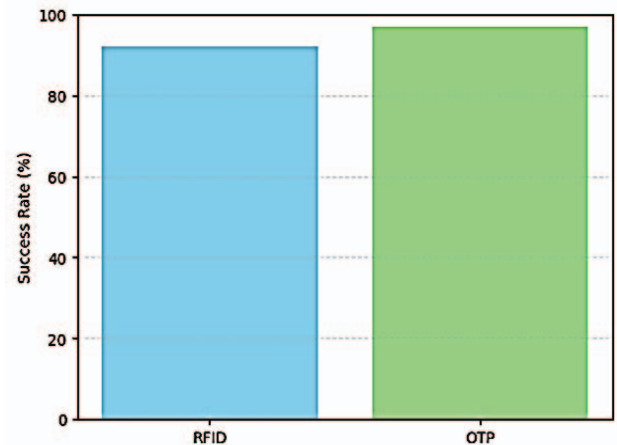


Fig.7: Authentication success rates.

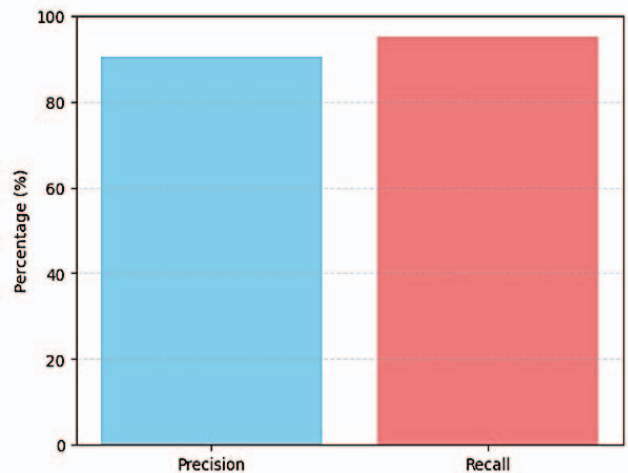


Fig.8: Precision and recall performance for intrusion detection.

unlocking ranges between 6–10 seconds, which is acceptable for real-time applications. Fig.9 presents the system’s performance in terms of the OTP-based access mechanism.

D. System Latency

The system latency refers to the time between user input and actuation of the door lock via servo motor. In RFID-based access, the average latency recorded from card detection to door unlocking has been recorded as 0.8 seconds, whereas OTP-based access exhibited a total latency of 1.2 seconds. It can be observed from Fig.10 that the system is fast enough for real time responses, ensuring smooth and responsive access control.

E. Cloud Communication Reliability

The Blynk's cloud platform plays an important role in delivering OTPs and transmitting the intrusion alerts. Over the course of 72 hours of testing, the system registered 98.5% server uptime to ensure the communication reliability. During the evaluation period, only three cloud message delivery failures were observed, all lasting less than 5 seconds. Such results validate the Blynk’s robustness in non-critical time-sensitive operations like OTP dispatch and notifications. Fig.11 depicts the reliability of the proposed system with cloud-based communications.

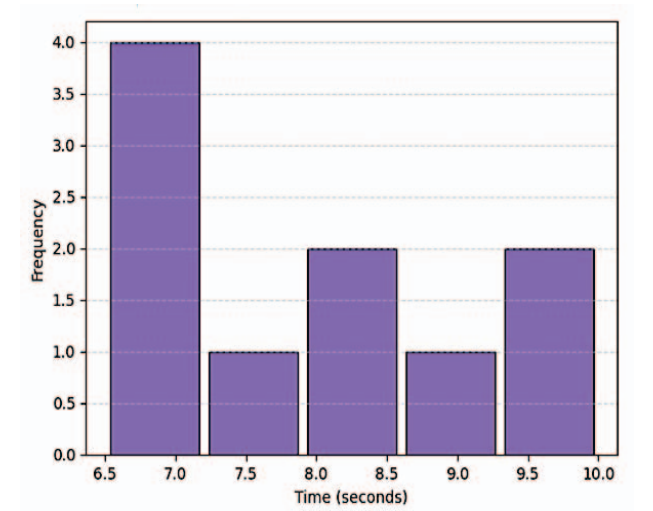


Fig.9: OTP delivery and authentication time.

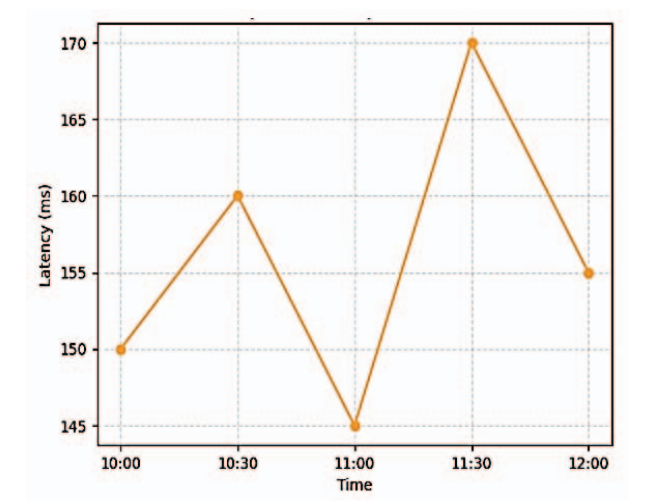


Fig.10: System latency over the operating intervals.

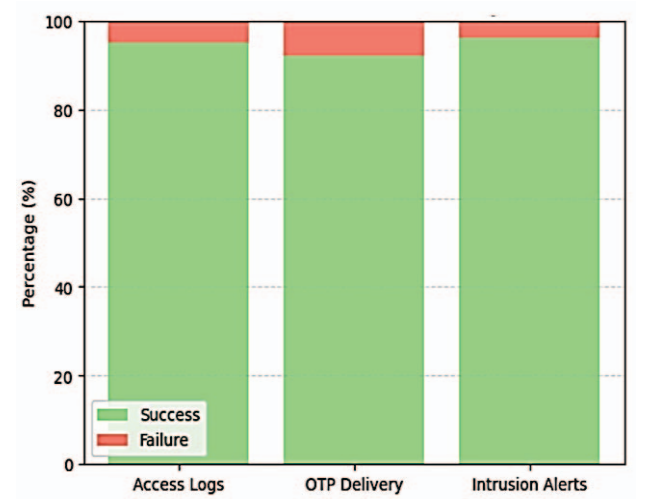


Fig.11: Cloud communication reliability.

A summary of various performance metrics of the proposed system has been presented in Table 1. From this table, it can be observed that the proposed system performs on a scale of above 90% in the areas of user authentication, intrusion detection and cloud uptime.

Based on a number of test cases, a summary of the expected and observed outputs of the proposed system has been presented in Table 2. It is observed that the proposed system performs up to the mark for the various possible cases of the RFID, OTP and intrusion detections.

Table 1: Summary of various performance metrics.

Metric	Value/Result
Authentication Success Rate	97%
Intrusion Detection Precision	90.47%
Intrusion Detection Recall	95%
Cloud Uptime (72 hrs)	98.5%
OTP-based Response Time	6–10 seconds
RFID Latency	0.8 seconds

Table 2: Summary of the expected and observed outputs.

Test Case	Expected Output	Observed Output
Valid RFID	Access Granted	Access Granted
Invalid RFID with valid OTP	Access Granted	Access Granted
Invalid RFID with invalid OTP	Access Denied	Access Denied
Intruder detected	Alert & Buzzer	Alert & Buzzer

V. CONCLUSION

This paper presents a smart door lock system that leverages RFID technology, OTP-based cloud verification, and motion detection to create a robust and intelligent access control mechanism. The use of the Blynk platform ensures real-time user engagement and notification, while the Wokwi simulation environment provides a reliable and accessible means to prototype and test the system. The system includes a real-time alert mechanism using both audible and remote notifications. If an unauthorized RFID tag is detected or if the OTP is entered incorrectly, the system triggers a buzzer alarm and simultaneously sends an SMS alert to a pre-configured admin number. This mechanism is critical for immediate threat response and system reliability. Upon successful dual verification with RFID and OTP, the microcontroller activates a relay circuit that drives the electronic unlocking system. After a predefined interval, the door automatically re-locks to maintain security. The proposed system offers an effective multi-factor authentication and breach alerts, making it suitable for residential buildings, offices, hostels, and restricted zones in the institutions. The proposed system can be made more scalable and efficient by incorporating some additional features such as voice control and AI-based processing for intrusion detections. Additionally, the GSM module can be used for SMS-based alerts in the areas with unreliable internet connectivity.

A comparison of the proposed work with the recently published works has been presented in Table 3. It can be observed that the proposed system performs better than the systems reported in [3], [5] and [13] in terms of the functionalities such as RFID, OTP, cloud service integration and intruder detection. However, voice control and Bluetooth features may be considered in the future scope of this work.

Table 3: Comparison of the proposed work with earlier published works.

Functionality	This Work	Ref. [3]	Ref. [5]	Ref. [13]
RFID Verification	Yes	No	No	Yes
OTP Verification	Yes	Yes	No	No
Cloud Service Integration	Yes	No	No	No
Intruder Detection	Yes	No	Yes	Yes
Voice Control	No	Yes	No	No
Bluetooth or LoRA	No	Yes	Yes	No

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